Riverline Audio Sourcing, Licensing & Architecture Specification

Independent Card-Room Foley & Licensing Audit + Extended Brief Synthesis for Milestone 3 (M3)

Prepared for: Riverline Owner & Core Engineering (`RIVERLINE_BRIEF.md` dated 2026-10-21) | **Date:** 2026-10-23

Executive Verdict ("Short Answer First")

[Judgement] Riverline can ship a complete, 100% real-recorded card-room foley and tactile study UI sound palette at **$0 cost** with full commercial clearance covering a free app and a later paid/supporter tier—**provided core shipped assets are restricted to CC0 1.0 and CC-BY 3.0/4.0 recordings from Kenney.nl, OpenGameArt.org, and Freesound.org**.

- **How `RIVERLINE_BRIEF.md` Extends This Research:** Incorporating the project brief transforms a generic sound search into an end-to-end engineering specification across seven dimensions: (1) **Section 1's open-vs.-closed source optionality** exposes a critical trap in Mixkit, Sonniss GDC, Zapsplat, and Pixabay (which permit compiled commercial games but forbid redistributing standalone audio files alongside source code); (2) **Principle #3 ("Heuristic disagreement is never a mistake") & Section 4 ("study cues never sound like right/wrong")** eliminate tonal success/failure chimes in favor of neutral physical latch/wood stops; and (3) **Section 4's 31-moment motion catalog & frequency rule** dictate exact cooldown and silence rules during rapid keyboard hand scrubbing.
- **Fully Cleared for Closed & Open Source (Sets 1, 2 & 3A):** Kenney.nl (CC0), OpenGameArt (CC0), and Freesound.org (CC0 & CC-BY) allow commercial use, EQ softening, and direct redistribution of raw `.ogg`/`.wav` files in both packaged Electron/browser builds and public Git repositories.
- **Cleared for Packaged Closed-Source Builds Only (Sets 3B & 4):** Sonniss #GameAudioGDC, Zapsplat, Mixkit, and Pixabay prohibit redistributing standalone audio files or shipping audio *"with source files"*.
- **Flagged & Disqualified:** BBC Sound Effects (RemArc Licence) and Freesound CC-BY-NC strictly forbid commercial use or revenue-generating supporter tiers.

Specification Scope	Core Sourcing & Legal Audit (Sections 2–4)	Extended Brief Synthesis (Sections 5–7)
Included Deliverables	Verbatim licence quotes for all 8 required sources; 4 coherent room/mic character sets; 65 verified candidates (5 per event across all 13 events) with direct URLs, specs, and redistribution verdicts.	7-dimension Brief impact analysis; 31-moment Motion setting coupling (`Instant/Faster/Standard/Reduced`); M2/M3 workspace trigger matrix; cross-platform codec plan (Firefox/Electron/iOS); offline `ffmpeg` mastering script & vanilla JS Web Audio module.

Epistemic Policy (`RIVERLINE_BRIEF.md §7`): Every cited page was opened and inspected. Verbatim licence clauses are quoted directly. Researcher inferences are marked **[Judgement]**. Sub-metrics not printed on a public web page (e.g., exact milliseconds inside ZIP packs) are marked `UNVERIFIED`.

1. How the Riverline Project Brief Reshapes & Extends the Research

[Judgement] Evaluating audio candidates without `RIVERLINE_BRIEF.md` would lead to three product errors: choosing royalty-free sounds that silently block future open-sourcing, picking bright "correct/wrong" chimes that violate Riverline's coaching philosophy, and triggering foley on every keypress during rapid hand-history review. The table below maps seven explicit rules from `RIVERLINE_BRIEF.md` directly to the extended engineering specifications in this report.

Riverline Brief Section	Verbatim Brief Requirement / Context	Direct Impact on Audio Sourcing, Licensing & Architecture
§1 Business & Source Model + §2 Principle #7	*"Mostly free, with an optional subscription / supporter tier / paid services / donations later… Source model (closed or open) is decided after the best tools' licences are known; currently treated as closed… Free first and always legal."*	**[Judgement]** Splits "free for commercial use" into two tiers: **Tier A (CC0 / CC-BY)** allows committing raw `.ogg`/`.wav` files directly into a public open-source Git repo; **Tier B (Sonniss GDC, Zapsplat, Mixkit, Pixabay)** forbids shipping standalone audio files or distributing *"with source files"*. Choosing Tier A preserves 100% source-model freedom.
§2 Principle #3 & §4 Sound Direction	*"Heuristic disagreement is never a 'mistake'. Comparisons are descriptive (inside / outside the baseline's mix, close)… only real, properly licensed recordings, judged by ear; study cues never sound like right/wrong."*	**[Judgement]** Disqualifies tonal game-show buzzers, major/minor "win/lose" jingles, and harsh error alerts (e.g., Kenney's `error_005.ogg`). `Confirm` must sound like a physical latch or wooden piece locking into place; `Gentle Error` must sound like a muted wooden mechanical stop (signaling an invalid UI action, never a strategic poker "blunder").
§2 Principle #1 & #2 (Honesty & Unknowns)	*"Confidence tiers: Exact, Verified, Estimated, Unavailable. No numeric confidence percentages, no fake GTO, no fake EV, no mastery theatre. Unknown is never fold, zero or excluded."*	**[Judgement]** Prohibits "mastery theatre" fanfare. Revealing an `Estimated` heuristic baseline or an `Unavailable` spot must never trigger celebratory or punitive sound effects; audio belongs strictly to physical table state changes, user commits, and calm session boundaries.
§4 Motion Catalog & Frequency Rule	*"Approved 'motion catalog' of 31 moments; frequency decides motion (keyboard and very frequent actions are instant)… one global Motion setting (Instant / Faster / Standard / Reduced); 60 fps."*	**[Judgement]** Requires coupling the audio engine to the global `Motion` setting and enforcing a `60–85 ms` cooldown / voice-stealing window. Rapid arrow-key hand-history scrubbing and keyboard range-matrix navigation must stay **silent** (just as their motion is `Instant`) to prevent machine-gun ear fatigue.
§1 First User Profile (Home Game vs. Online)	*"First user: the owner (home games 6–8 handed with 30–60 bb buy-ins and no rake; online 9-max at 100 bb+; studies a lot)."*	**[Judgement]** In 30–60 bb home games, `All-In Push` and 3-bet stack commits occur much more often than at 200 bb. Chip cues must use genuine 11.5g–14g clay/ceramic home-game chips on wood/felt (SilverDubloons, farthestar, Benjamin152) rather than hollow plastic or noisy casino slot cascades.
§4 Look & Mascot (Forest / Jade / Brass + Chippy)	*"Look: dark forest / jade / brass, 'calm surface, deep data'… mascot Chippy (M3) is a calm teacher-like poker chip."*	**[Judgement]** Translates visual materials into acoustic materials: damped baize/wood for table actions (dark forest), solid clay/ceramic tokens for bets and Chippy's M3 micro-cues (jade), and warm real brass (loganzsound RX-denoised brass chimes) reserved for `Session Finish`.

Riverline Brief Section	Verbatim Brief Requirement / Context	Direct Impact on Audio Sourcing, Licensing & Architecture
§1 & §3 Tech Stack (Plain JS, Firefox, Electron, Phone)	*"Plain JavaScript (no React/Vue/Tailwind): one large classic script `app/src/core/logic.js` (~14k lines) plus ES modules… Firefox on Windows… thin Electron host; a phone version is planned."*	**[Judgement]** Requires a zero-dependency ES module (`app/src/core/audio.js`) using native Web Audio API `AudioContext`, pre-decoded buffers, iOS Safari touch-unlock handling, and a dual-codec container strategy (`.ogg` primary + `.m4a`/`.wav` fallback for iOS WebKit).

2. Source-by-Source Licence Audit (All 8 Required Sources Opened & Quoted)

Every required source below was opened and audited against Riverline's business and distribution model: free application with a possible paid/supporter tier later, zero non-commercial restrictions, and full transparency on whether raw audio files can be redistributed inside the app bundle and repository.

Source & Licence Link	Verbatim Quoted Licence Clause(s)	Paid Tier & App Redistribution Verdict
1. Kenney.nl CC0 1.0 Universal (OGA Mirror)	*"License: Creative Commons CC0" / "Attribution Instructions: Credit 'Kenney.nl' or 'www.kenney.nl', this is not mandatory."*	**Paid Tier: YES** (No credit required) **Redistrib: YES (Unrestricted)** — Cleared for closed builds, web bundles, and public open-source Git repos.
2. Freesound.org CC0 1.0 & CC-BY 4.0 / 3.0 *(Exclude CC-BY-NC)*	**CC0:** *"Creative Commons 0 — You can copy, modify, distribute and perform the sound, even for commercial purposes, all without the need of asking permission to the author."* **CC-BY:** *"Attribution 4.0 — You are free to share (to copy, distribute and transmit) and to remix (to adapt and modify) as long as you credit the author of the sound."*	**Paid Tier: YES** for CC0 & CC-BY (**NO** for CC-BY-NC) **Redistrib: YES (Unrestricted)** — CC0 needs zero credit; CC-BY needs a simple credit line in Riverline's About panel.
3. OpenGameArt CC0 1.0 Universal (80 CC0 RPG SFX)	*"License(s): CC0" / "Sound effects with attribution not required"*	**Paid Tier: YES** (No credit required) **Redistrib: YES (Unrestricted)** — Cleared for closed builds, web bundles, and public open-source repos.
4. Sonniss GDC Bundles Sonniss #GameAudioGDC License v2.0 (Archive)	*"a) Licensee may use the licensed sound effects on an unlimited number of projects for the entirety of their life time. b) Licensee may use and modify the licensed sound effects for personal and commercial projects without attribution to the original creator. c) Licensee may publicly perform a reproduction of the sound effects over any form of medium, as long as they are synchronized within the Licensee's project… This right does not extend to supplying the sound effects themselves, whether as files or as the output of a device or service… This does not restrict the Licensee's finished projects, which may be shared, published and sold in the normal way."*	**Paid Tier: YES** (No credit required; McHugh's pack is in GDC Part 2 or $25 standalone) **Redistrib: CONDITIONAL (Packaged App Only)** — **[Judgement]** Clause (c) prohibits supplying the files themselves, making public open-source Git repo redistribution risky.

Source & Licence Link	Verbatim Quoted Licence Clause(s)	Paid Tier & App Redistribution Verdict
5. BBC Sound Effects BBC Content Licence for RemArc (Catalogue)	*"7. Using BBC content: Provided you keep to these rules, the BBC grants you permission to use the BBC content but only... For non-commercial, personal or research purposes... 8. Using BBC content for business: a) When you need permission: If you're intending to use it for any other purpose, for example... substantially to do your job – as an employee, contractor or consultant; for commercial purposes – to make a profit; for non-profit and government organisations... you'll need to get our permission first, and you might have to pay a fee. ... Pro Sound Effects operate the commercial licensing of BBC sound effects."*	**Paid Tier: NO — FLAGGED & DISQUALIFIED** **Redistrib: NO** — Strictly personal, educational, and non-commercial under RemArc. Cannot be used in an app with a paid/supporter tier without paid commercial licensing.
6. Zapsplat Zapsplat Standard License	*"2. Basic (Free) Users: Usage Rights: You may download and use our sound effects and music tracks in an unlimited number of personal, commercial, and broadcast projects worldwide, in perpetuity... Attribution Required: You must provide clear credit to 'ZapSplat'... 4. Permitted Uses: ...embedded or synchronized within a production, in any project including... Games, apps, and software. ... 5. Prohibited Uses: You may not, under any circumstances: ... Redistribute our sounds in any form... 6. ...provided the Licensed Audio is incorporated into a larger end product and is not distributed as standalone audio files."*	**Paid Tier: YES** (Free tier = MP3 + credit; £7.99/mo or £69.99/yr = WAV + no credit) **Redistrib: CONDITIONAL (Embedded in Packaged App Only)** — Cannot be redistributed as standalone files in an open-source repo.
7. Mixkit Mixkit Sound Effects Free License	*"Items under the Mixkit Sound Effects Free License can be used in your commercial and non-commercial projects for free. You are licensed to use the Item to create an End Product that incorporates the Item as well as other things, so that it is larger in scope and different in nature than the Item. You're permitted to download, copy, modify, distribute and publicly perform the Sound Effect Items on any web or social media platform, in podcasts and in video games... You can't redistribute the Item on its own, as stock, in a tool or template, or with source files."*	**Paid Tier: YES** (No credit required) **Redistrib: CONDITIONAL (Closed-Source Only) — [Judgement]** Explicitly states *"You can't redistribute the Item... with source files"*, barring open-source Git distribution.
8. Pixabay Sounds Pixabay Content License (Legal Extract, Audit)	*"Under Pixabay's terms, you may not sell or distribute content (whether in digital or physical format) on a stand-alone basis (i.e., where no creative effort has been applied to the content and it remains substantially in the same form as it exists on the Pixabay website). As long as you have created a unique and original creative work, the Pixabay license allows our community to use your work for both non-commercial and commercial purposes. ... No attribution or revenue sharing is required..."*	**Paid Tier: CONDITIONAL** (Abandoned CC0 on Jan 1, 2019; $0 indemnification) **Redistrib: HIGH RISK — [Judgement]** Prohibits standalone digital distribution and suffers from unvetted user re-uploads.

3. Coherent Sound Sets (Grouped by Room, Mic & Material Character)

[Judgement] Combining bright condenser recordings on bare hardwood with dark dynamic-mic recordings on padded baize makes a table interface sound disjointed. The 65 verified candidates in Section 4 are organized

into **four coherent sets** so you can adopt a unified acoustic environment:

- **Set 1: "Dry Studio Tabletop" — Kenney CC0 Unified Suite (Zero-Friction CC0 Baseline):** Uses Kenney Casino Audio (OGA Mirror, File Directory), Kenney Impact Sounds, Kenney UI Audio (File Directory), and Kenney Interface Sounds (File Directory). Recorded in a dry studio at `44.1 kHz` Stereo OGG Vorbis with physical cards, composite chips, damped wooden blocks, and switches, pre-trimmed to `0.03s–0.45s`. Apply a `14 kHz` low-pass filter to remove light airy hiss noted on raw takes.
- **Set 2: "Close-Mic Home Game Foley" — Curated Freesound CC0 & CC-BY Studio Set (Highest Realism):** Uses uncompressed WAV foley on Freesound.org by SilverDubloons (Blue Yeti clay chips & cards), el_boss, filmfan87 (`48 kHz / 24-bit` card placement), uEffects (AT2020 card session), f4ngy (CC-BY 4.0 card flip), farthestar (real clay/ceramic chips), Benjamin152 (Zoom H4n 24-bit stereo chips & wood, CC-BY 3.0), Anthousai (Tascam DR-40 96 kHz wood tap), Churrumaki (real fingers tapping table), SignatureSoundsOrg (96 kHz pistachio-shell UI clicks), and loganzsound (96 kHz AKG C414 XLII switches & RX 11 denoised homemade brass chimes).
- **Set 3: "High-Res 96 kHz Mono Foley Studio" — Sonniss #GameAudioGDC + OpenGameArt CC0 Packs:** Pairs Timothy McHugh's Playing Cards & Betting Chips (`96 kHz / 24-bit Mono`, 174 WAVs in Sonniss #GameAudioGDC Part 2 under the GDC Bundle License, designated **Set 3B** for packaged builds) with `HaelDB`'s CC0 Card Game Sounds (made for Cockatrice) and `rubberduck`'s CC0 80 CC0 RPG SFX (designated **Set 3A**, unrestricted CC0).
- **Set 4: "Commercial Felt-Table & Wooden Mallet Alternates" — Zapsplat & Mixkit (Packaged Builds Only):** Combines Zapsplat Felt-Table Cards & Chips and Zapsplat Wooden Mallet Tones with Mixkit Casino, Typewriter, Click, and Notification SFX (Verified Asset Index).

4. Per-Event Candidate Evidence Tables (All 13 Required Events, 5 Candidates Each)

Event 1: Card Dealt / Slid

Target Character: Crisp, brief friction of a single card sliding off the deck across cloth/wood (`~180–320 ms`), never a noisy riffle shuffle.

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. `cardSlide1.ogg–cardSlide4.ogg` (Kenney Casino Audio / Index)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	`~0.25s` (`16.8–20.5 KB`, ms `UNVERIFIED`) OGG `44.1 kHz` Stereo	Four round-robin variations of a single playing card sliding briskly across a dry tabletop.
2. `Playing Card Slide Right` (Freesound #571581)	**Set 2** `el_boss`	*"Creative Commons 0 — You can copy, modify, distribute and perform..."* **Redistrib: YES** (CC0)	`0:00.265` (`46.0 KB`) WAV `44.1 kHz` 16-bit Mono	Clean, dry close-mic recording of a single playing card sliding to the right with a soft stop.
3. `slideCard04` (Freesound #817579)	**Set 2** `SilverDubloons`	*"Creative Commons 0"* **Redistrib: YES** (Unrestricted CC0)	`UNVERIFIED` (`<0.5s`) WAV / Blue Yeti (`UNVERIFIED` kHz)	Real plastic-coated playing card sliding smoothly off the top of another card.

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
4. `Draw Card (Cardsounds.zip)` (OGA Card Game Sounds)	**Set 3A** HaelDB	*"License(s): CC0"* **Redistrib: YES** (Unrestricted CC0)	`UNVERIFIED` (`<0.4s` in `238.2 KB` ZIP) WAV/OGG (`UNVERIFIED` kHz)	Real tabletop card draw friction recorded specifically for the open-source *Cockatrice* card client.
5. `Paper slide` (#1530) (Mixkit Casino / Index)	**Set 4** Mixkit	*"...commercial and non-commercial projects for free... can't redistribute... with source files."* **Redistrib: CONDITIONAL** (Closed-source)	`0:01.0` WAV / MP3 (`UNVERIFIED` kHz)	Smooth, softened friction of stiff cardstock gliding across a flat tabletop surface.

Event 2: Board Card Placed

Target Character: Slightly firmer, grounded contact of a community card settling flat onto the felt (`~180–350 ms`).

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. `cardPlace1.ogg`–`cardPlace4.ogg` (Kenney Casino Audio / Index)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	`~0.25s` (`17.1–25.7 KB`, ms `UNVERIFIED`) OGG `44.1 kHz` Stereo	Four distinct round-robin recordings of a playing card being set firmly onto a tabletop.
2. `card placed on table.wav` (Freesound #108395)	**Set 2** `filmfan87`	*"Creative Commons 0 — You can copy, modify, distribute and perform..."* **Redistrib: YES** (CC0)	`0:13.546` (Multi-take; slice `~0.25s` hits) WAV `48 kHz` 24-bit Mono	High-resolution mono foley session containing multiple clean takes of a single card settling on a table.
3. `Play card (Cardsounds.zip)` (OGA Card Game Sounds)	**Set 3A** HaelDB	*"License(s): CC0"* **Redistrib: YES** (Unrestricted CC0)	`UNVERIFIED` (`<0.4s` in `238.2 KB` ZIP) WAV/OGG (`UNVERIFIED` kHz)	Direct, lightly damped card placement thud recorded for tabletop card play.
4. `Playing Cards & Betting Chips` (Card Place takes) (Sonniss McHugh / License)	**Set 3B** Timothy McHugh	*"Licensee may use and modify... without attribution... synchronized within the Licensee's project."* **Redistrib: CONDITIONAL** (Packaged app)	`UNVERIFIED` per take (`174` files, `86.2 MB`) WAV `96 kHz` 24-bit Mono	Studio-grade 96 kHz mono foley of individual playing cards placed cleanly onto felt and wood.

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
5. `Poker card placement` (Mixkit Casino SFX)	**Set 4** Mixkit	*"...commercial and non-commercial projects for free... can't redistribute... with source files."* **Redistrib: CONDITIONAL** (Closed-source)	`0:01.0` WAV / MP3 (`UNVERIFIED` kHz)	Crisp single poker card placement on a damped card table surface.

Event 3: Card Flip

Target Character: Quick, light finger-snap lift and turn of hole cards at showdown or during hand replay (`~180–380 ms`).

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. `cardTakeOutPackage1.ogg` & `cardFan1.ogg` (Kenney Casino Audio / Index)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	`~0.25–0.40s` (`17.1 KB` & `32.2 KB`) OGG `44.1 kHz` Stereo	Snappy card-edge lift and turn that pairs seamlessly with `cardSlide*` and `cardPlace*`.
2. `Card Flip` (Freesound #240776)	**Set 2** `f4ngy`	*"Attribution 4.0 — You are free to share... and to remix... as long as you credit the author..."* **Redistrib: YES** (CC-BY 4.0; credit `f4ngy`)	`0:00.589` (`101.6 KB`) WAV `44.1 kHz` 16-bit Stereo	Natural home-recorded playing card flip with a crisp paper inflection and soft landing.
3. `Cards Sounds (AT2020 Foley Session)` (Freesound #208790)	**Set 2** `uEffects`	*"Creative Commons 0 — You can copy, modify, distribute and perform..."* **Redistrib: YES** (CC0)	`0:20.562` (Multi-take; slice `~0.25s` flips) WAV `44.1 kHz` 16-bit Mono	Warm Audio-Technica AT2020 condenser recording of single card turns and flips (*"Not too snappy"*).
4. `Tap / Untap (Cardsounds.zip)` (OGA Card Game Sounds)	**Set 3A** `HaelDB`	*"License(s): CC0"* **Redistrib: YES** (Unrestricted CC0)	`UNVERIFIED` (`<0.35s` in `238.2 KB` ZIP) WAV/OGG (`UNVERIFIED` kHz)	Subtle tabletop card rotation/flip foley designed for frequent card-state toggles.
5. `Poker card flick` & `Page turn single` (#1104) (Mixkit Casino / Index)	**Set 4** Mixkit	*"...commercial and non-commercial projects for free... can't redistribute... with source files."* **Redistrib: CONDITIONAL** (Closed-source)	`0:01.0` WAV / MP3 (`UNVERIFIED` kHz)	Quick fingernail flick and turnover of a stiff playing card.

Event 4: Check Knock on Felt

Target Character: Damped double-tap of knuckles or fingers on padded felt over wood (~200–450 ms), avoiding hollow door-slam resonances.

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. impactSoft_medium_000.ogg–004.ogg & impactWood_light_000.ogg (Kenney Impact Sounds)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	~0.18–0.35s (UNVERIFIED exact ms) OGG 44.1 kHz Stereo	Damped soft surface thud that can be triggered as a single tap or rapid 110 ms double-tap for a felt check.
2. dedotoca mesa / fingers touching table .wav (Freesound #563818)	**Set 2** Churrumaki	*"Creative Commons 0 — You can copy, modify, distribute and perform..."* **Redistrib: YES** (CC0)	0:04.565 (Slice 0.35s double-tap) WAV 48 kHz 16-bit Stereo	Authentic close-mic recording of human fingers tapping gently twice on a tabletop surface.
3. hit - wooden 07.wav (Freesound #406275)	**Set 2** Anthousai	*"Creative Commons 0"* **Redistrib: YES** (Unrestricted CC0)	0:00.551 (310.2 KB) WAV 96 kHz 24-bit Stereo	Warm, muted wooden tap recorded on a Tascam DR-40 that sounds like a single knuckle check on a table rail.
4. Knocking on Wood Door (1) (#629987) & knocking sound (#202231)	**Set 2** Flem0527 / deraj	*"Creative Commons 0"* (both) **Redistrib: YES** (Unrestricted CC0)	0:00.704 / 0:02.519 WAV 44.1 kHz 24-bit / 16-bit	Natural double knuckle knock on solid wood (apply a 1.5 kHz high-shelf cut to simulate felt padding over wood).
5. Wooden box on wooden table & Wooden box lid soft slam (Freesound Benjamin152)	**Set 2** Benjamin152	*"Attribution 3.0"* (CC-BY 3.0) **Redistrib: YES** (Credit Benjamin152)	0:07.795 / 0:12.0 (Slice 0.3s thud) WAV 44.1 kHz 24-bit Stereo	Soft, resonant wood-on-wood percussion recorded with a Zoom H4n at 30 cm distance.

Event 5: Chips Bet (Small / Medium / Large Stacks)

Target Character: Real clay/ceramic poker chips (never hollow plastic or slot-machine coins) scaled across three distinct stack weights: 1–2 chips (small), 3–5 chips (medium), and 6–10+ chips (large).

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. Small: chipLay1–3.ogg | Med: chipsCollide1–4.ogg | Large: chipsStack1–6.ogg (Kenney Casino Audio)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	~0.15–0.35s (10.5–14.1 KB) OGG 44.1 kHz Stereo	Cohesive 13-file graduated set covering single chip lays, light clinks, and firmer multi-chip stack placements.
2. Small: chip05 (#817556) & Large: clatter10 (#817567)	**Set 2** SilverDubloons	*"Creative Commons 0 — You can copy, modify, distribute and perform..."* **Redistrib: YES** (CC0)	0:00.216 (chip05) / UNVERIFIED WAV 22.05 kHz 16-bit Stereo	Real clay poker chip dropped onto another (chip05) and a handful of clay chips clattering onto wood (clatter10).

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
3. Med/Large: `poker_chips6.wav` (#201808) & Small/Med series	**Set 2** `farthestst ar`	*"Creative Commons 0"* **Redistrib: YES** (Unrestricted CC0)	`UNVERIFIED (~0.4–1.0s)` WAV (`UNVERIFIED` exact kHz)	Authentic clay/ceramic poker chips clinking together (*"from clay/ceramic poker chips, not the cheap plastic ones"*).
4. Small/Med/Large: `Poker Chips landing on a wooden Table` (#445141) & `stacking` (Profile)	**Set 2** `Benjamin15 2`	*"Attribution 3.0 — You are free to share... and to remix..."* **Redistrib: YES** (CC-BY 3.0; credit `Benjamin152`)	`0:17.252` / `0:11.0` (Multi-take) WAV `44.1` kHz 24-bit Stereo	Multi-take stereo Zoom H4n session of real poker chips stacking and landing in small, medium, and large clusters.
5. `Playing Cards & Betting Chips` (Sonniss) & `Poker chips x8 on felt table` (Zapsplat)	**Set 3B / 4** T. McHugh / ZapSplat	Sonniss GDC / Zapsplat Standard **Redistrib: CONDITIONAL** (Packaged app)	`UNVERIFIED` / `~0:01.0` WAV `96` kHz 24-bit / MP3	Studio foley of single, medium, and 8-chip stacks set down directly on a wooden felt-covered poker table.

Event 6: All-In Push

Target Character: Deliberate, weighty forward slide and settle of a heavy stack of clay chips across felt (`~500–950 ms`)—decisive, never theatrical or accompanied by sirens/horns.

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. `chipsHandle5.ogg` (`26.8 KB`) + `cardShove2.ogg` (`29.4 KB`) (Kenney Casino Audio / Index)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	`~0.60–0.85s` (`26.8 KB` + `29.4 KB`) OGG `44.1` kHz Stereo	Sustained multi-chip handling cascade (`chipsHandle5`) layered with the low-end felt friction of `cardShove2`.
2. `clatter10` (Freesound #817567)	**Set 2** `SilverDubl oons`	*"Creative Commons 0"* **Redistrib: YES** (Unrestricted CC0)	`UNVERIFIED (~0.8–1.1s)` WAV / Blue Yeti (`UNVERIFIED` kHz)	Heavy handful of real clay poker chips pushed and settling onto a wooden tabletop with natural stack collapse.
3. `Poker Chips landing on a wooden Table` (Heavy Push Take) (Freesound #445141)	**Set 2** `Benjamin15 2`	*"Attribution 3.0"* (CC-BY 3.0) **Redistrib: YES** (Credit `Benjamin152`)	`0:17.252` total (Slice `~0.85s` take) WAV `44.1` kHz 24-bit Stereo	Dense, realistic multi-chip forward cascade and settle recorded in 24-bit stereo on a Zoom H4n.
4. `Handfull of Poker Chips` (Freesound Za-Games Profile)	**Set 2** `Za-Games`	*"Creative Commons 0"* **Redistrib: YES** (Unrestricted CC0)	`0:00.352` WAV `44.1` kHz 16-bit Mono	Compact, punchy release of a full handful of poker chips into the center pot.

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
5. `Playing Cards & Betting Chips` (All-In Stack Push takes) (Sonniss McHugh / License)	**Set 3B** Timothy McHugh	*"Licensee may use and modify... without attribution... synchronized within the Licensee's project."* **Redistrib: CONDITIONAL** (Packaged app)	`UNVERIFIED` per take (174 files total) WAV `96 kHz` 24-bit Mono	Dedicated close-mic foley of heavy chip stacks pushed across cloth and collapsing softly into the pot.

Event 7: Pot Pushed to Winner

Target Character: Warm, unhurried sweep and gathering clatter of chips being scooped across the felt toward the winning seat (`~600–1,150 ms`).

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. `chipsHandle6.ogg` (18.9 KB) / `chipsHandle1.ogg` (20.8 KB) + `chipsCollide3.ogg` (Kenney Casino Audio)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	`~0.55–0.80s` (18.9–20.8 KB) OGG `44.1 kHz` Stereo	Smooth multi-chip gathering shuffle that evokes hands sweeping a pot across tabletop cloth.
2. `Poker Chips stacking & landing on a wooden Table` (Freesound #445141 & Profile)	**Set 2** `Benjamin152`	*"Attribution 3.0"* (CC-BY 3.0) **Redistrib: YES** (Credit `Benjamin152`)	`0:11.0` / `0:17.252` (Slice `~0.95s` sweep) WAV `44.1 kHz` 24-bit Stereo	Natural stereo movement of multiple clay chips being gathered, slid, and stacked together on wood.
3. `poker_chips6.wav` (Freesound #201808)	**Set 2** `farthestst ar`	*"Creative Commons 0"* **Redistrib: YES** (Unrestricted CC0)	`UNVERIFIED` (`~0.7–1.0s`) WAV (`UNVERIFIED` exact kHz)	Rich, warm ceramic/clay chip gathering clink without harsh metallic ring.
4. `Playing Cards & Betting Chips` (Pot Sweep takes) (Sonniss McHugh / License)	**Set 3B** Timothy McHugh	*"Licensee may use and modify... without attribution... synchronized within the Licensee's project."* **Redistrib: CONDITIONAL** (Packaged app)	`UNVERIFIED` per take (174 files total) WAV `96 kHz` 24-bit Mono	High-definition foley of a dealer/player sweeping a pile of betting chips across felt.
5. `21x item (stone, wood) (80-CC0-RPG-SFX.zip)` (OGA 80 CC0 RPG SFX)	**Set 3A** `rubberduck`	*"License(s): CC0"* **Redistrib: YES** (Unrestricted CC0)	`UNVERIFIED` (`1.8 MB` ZIP archive) WAV/OGG (`UNVERIFIED` kHz)	Real-recorded stone/wood token gathering clinks (use muted stone/wood takes, avoiding metallic coins).

Event 8: Fold (Cards Tossed)

Target Character: Soft, dismissive forward flick/shove of two hole cards sliding into the muck (~180–350 ms), lower in pitch and softer than an active deal.

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. cardShove1.ogg–cardShove4.ogg (Kenney Casino Audio / Index)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	~0.25–0.35s (23.2–29.4 KB) OGG 44.1 kHz Stereo	Purpose-recorded card-mucking shove (cardShove1–4) with a heavier, softer cloth deceleration than cardSlide.
2. Cards Sounds (Soft Muck / Toss Slice) (Freesound #208790)	**Set 2** uEffects	*"Creative Commons 0 — You can copy, modify, distribute and perform..."* **Redistrib: YES** (CC0)	0:20.562 total (Slice ~0.28s toss) WAV 44.1 kHz 16-bit Mono	Soft, unhurried two-card toss onto a tabletop recorded on an Audio-Technica AT2020 condenser.
3. Playing Card Slide Right (Pitch -2 st, 10 kHz LPF) (Freesound #571581)	**Set 2** el_boss	*"Creative Commons 0"* **Redistrib: YES** (Unrestricted CC0)	0:00.265 (46.0 KB) WAV 44.1 kHz 16-bit Mono	Quick card slide across table that becomes an unmistakable muted muck/fold when darkened against the deal cue.
4. 4x book/page flip (80-CC0-RPG-SFX.zip) (OGA 80 CC0 RPG SFX)	**Set 3A** rubberduck	*"License(s): CC0"* **Redistrib: YES** (Unrestricted CC0)	UNVERIFIED (<0.4s in 1.8 MB ZIP) WAV/OGG (UNVERIFIED kHz)	Soft paper-on-paper brush that works well as a quiet, non-intrusive fold cue during rapid hand review.
5. A handful of playing cards drop on table 5 (Zapsplat Cards Page)	**Set 4** ZapSplat	*"You may download and use... credit to 'ZapSplat'... may not... Redistribute..."* **Redistrib: CONDITIONAL** (Packaged app)	0:01.0 MP3 (Free) / WAV (Paid)	Light drop/toss of playing cards settling flat onto a tabletop.

Event 9: Soft UI Tick

Target Character: Ultra-short, dry organic micro-click (25–70 ms) for secondary UI interactions, sizing sliders, or filter toggles (suppressed on rapid keyboard scrubbing per RIVERLINE_BRIEF.md §4).

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. click1.ogg (8.8 KB), rollover1.ogg (11.0 KB), or click_001.ogg (6.8 KB) (Kenney UI & Interface Sounds)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	~0.03–0.07s (6.8–11.0 KB) OGG 44.1 kHz Stereo	Featherweight, dry micro-ticks that add tactile presence without tonal pitch or ear fatigue.

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
2. `UI_Game_Clicks_From_Pistachio_Perc_19` (#869735) & `_27` (#869743)	**Set 2** `SignatureSoundsOrg`	*"Creative Commons 0 — You can copy, modify, distribute and perform…"* **Redistrib: YES** (CC0)	`0:01.837` (Trim single hit to ~`0.04s`) WAV `96 kHz` 16-bit Stereo	Real pistachio-shell percussion shaped into dry, wood-like organic UI ticks (*"organic tactile quality"*).
3. `Tic Toc Click` & `Normal click` (Freesound Pack #25371)	**Set 2** `Breviceps`	*"Creative Commons 0"* **Redistrib: YES** (Unrestricted CC0)	`0:00.031` / `0:00.048` WAV / FLAC (`UNVERIFIED` kHz)	Ultra-tight `31 ms` and `48 ms` physical clicks ready for zero-latency UI feedback without trimming.
4. `Hover over Or interface click (Cardsounds.zip)` (OGA Card Game Sounds)	**Set 3A** `HaelDB`	*"License(s): CC0"* **Redistrib: YES** (Unrestricted CC0)	`UNVERIFIED` (<`0.08s` in `238.2 KB` ZIP) WAV/OGG (`UNVERIFIED` kHz)	Subtle tabletop card-client hover/click cue designed specifically to accompany card deal/place sounds.
5. `Typewriter soft hit (#1366)` & `Select click (#1109)` (Mixkit Typewriter / Index)	**Set 4** Mixkit	*"…commercial and non-commercial projects for free… can't redistribute… with source files."* **Redistrib: CONDITIONAL** (Closed-source)	`0:00.5` / `0:01.0` WAV / MP3 (`UNVERIFIED` kHz)	Real damped mechanical key-cap tap (#1366) with a warm, study-desk character.

Event 10: Confirm

Target Character: Grounded, reassuring physical switch or magnetic/wooden lock-in (`~90–220 ms`)—affirming an action or saving a Personal Strategy node without sounding like a game-show "correct answer!" chime (`RIVERLINE_BRIEF.md §2 & §4`).

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. `drop_001.ogg` (`7.75 KB`) or `switch1.ogg` (`12.97 KB`) (Kenney Interface & UI Audio)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	`~0.10–0.18s` (`7.8–13.0 KB`) OGG `44.1 kHz` Stereo	Warm, wooden/tactile pop-drop (`drop_001`) or decisive physical toggle (`switch1`) without judgmental pitch.
2. `Deep Click` (Freesound #874468)	**Set 2** `loganzsound`	*"Creative Commons 0 — You can copy, modify, distribute and perform…"* **Redistrib: YES** (CC0)	`0:00.190` (`54.2 KB`) WAV `96 kHz` 24-bit Mono	Rich, low-mid physical switch engagement (`190 ms`) with a satisfying mechanical weight.
3. `AirPods Pro 2 Case Flick Open Close 3x (Single Close Snap)` (Freesound #871536)	**Set 2** `loganzsound`	*"Creative Commons 0"* **Redistrib: YES** (Unrestricted CC0)	`0:02.991` total (Slice ~`0.15s` snap) WAV `96 kHz` 24-bit Mono	Ultra-clean, damped magnetic latch snap recorded on an AKG C414 XLII that feels like locking a strategy tile into place.

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
4. `Button Click` (100% Real Foley) (Freesound #858919)	**Set 2** `DeWildAshDev`	*"Creative Commons 0"* **Redistrib: YES** (Unrestricted CC0)	`0:00.847` (Trim tail to `~0.20s`) WAV `44.1 kHz` 16-bit Stereo	Crisp two-part physical object click explicitly documented as *"recorded using everyday objects... No AI involved"*.
5. `On or off light switch tap` (#2585) (Mixkit Click SFX / Index)	**Set 4** Mixkit	*"...commercial and non-commercial projects for free... can't redistribute... with source files."* **Redistrib: CONDITIONAL** (Closed-source)	`0:01.0` WAV / MP3 (`UNVERIFIED` kHz)	Solid, familiar physical rocker-switch tap for committing a hand or filter state.

Event 11: Gentle Error

Target Character: Soft, low muted wooden bump or blocked mechanical key (`~110–250 ms`) signaling an invalid UI input (e.g. duplicate card selection or impossible stack size)—strictly never a punitive "wrong answer" buzzer (`RIVERLINE_BRIEF.md §2 Principle #3 & §4`).

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. `impactSoft_heavy_000.ogg` (Kenney Impact) or `back_001.ogg` (`8.9 KB`, Interface)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	`~0.15–0.24s` (`8.88 KB` for `back_001`) OGG `44.1 kHz` Stereo	Muted, low-frequency cushioned bump (`impactSoft_heavy_000`) or understated downward wooden tap (`back_001`).
2. `hit - wooden 07.wav` (Low-passed at `2.5 kHz`) (Freesound #406275)	**Set 2** `Anthousai`	*"Creative Commons 0 — You can copy, modify, distribute and perform..."* **Redistrib: YES** (CC0)	`0:00.551` (Trim to `0.22s`) WAV `96 kHz` 24-bit Stereo	Damped, hollow wooden stop recorded on a Tascam DR-40 that communicates "boundary reached" physically.
3. `billiard ball clack` (Low-passed & softened) (Freesound #539854)	**Set 2** `Za-Games`	*"Creative Commons 0"* **Redistrib: YES** (Unrestricted CC0)	`0:00.239` (`20.6 KB`) WAV `44.1 kHz` 16-bit Mono	Short, dry phenolic resin knock (`239 ms`) that feels like two solid tokens bumping when an invalid card is clicked.
4. `Error` (OGA Card Game Sounds) & `5x wood hits` (80 CC0 RPG SFX)	**Set 3A** `HaelDB` / `rubberduck`	*"License(s): CC0"* (both) **Redistrib: YES** (Unrestricted CC0)	`UNVERIFIED` (`<0.30s` in ZIP archives) WAV/OGG (`UNVERIFIED` kHz)	Tabletop card-client blocked-action cue (`HaelDB`) and dry muted wood hits (`rubberduck`).

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
5. `Mechanical typewriter single hit` (#1382) (Mixkit Typewriter / Index)	**Set 4** Mixkit	*"...commercial and non-commercial projects for free... can't redistribute... with source files."* **Redistrib: CONDITIONAL** (Closed-source)	`0:01.0` WAV / MP3 (`UNVERIFIED` kHz)	Tactile mechanical key bottoming out against a physical stop bar without producing a bell.

Event 12: Session Finish (Short, Warm)

Target Character: Short, warm acoustic resonance (`~650–1,400 ms`)—such as a real brass chime, desk bell, or wooden marimba mallet—matching Riverline's "dark forest / jade / brass" palette (`RIVERLINE_BRIEF.md §4`).

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. `impactBell_heavy_000.ogg` / `003.ogg` (Kenney Impact) or `bong_001.ogg` (`6.8 KB`, Interface)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	`~0.55–1.10s` (`6.77 KB` for `bong_001`) OGG `44.1 kHz` Stereo	Warm, mellow acoustic bell or wooden-bong strike with a clean, unhurried decay.
2. `Chime Ball Pendant Sparkle Shakes Short 01` (#873724) & `Brass Wind Chimes` (Pack #46478)	**Set 2** `loganzsound`	*"Creative Commons 0 — You can copy, modify, distribute and perform..."* **Redistrib: YES** (CC0)	`0:11.487` total (Slice `1.1s` brass note) WAV `48 kHz` 24-bit Mono	Real close-mic brass and silver chime recordings (RX 11 denoised) with a calm metallic warmth matching Riverline's brass accents.
3. `Stage Change` / `Game Start` (`Cardsounds.zip`) (OGA Card Game Sounds)	**Set 3A** `HaelDB`	*"License(s): CC0"* **Redistrib: YES** (Unrestricted CC0)	`UNVERIFIED` (`~0.6–1.2s` in `238.2 KB` ZIP) WAV/OGG (`UNVERIFIED` kHz)	Understated tabletop session boundary cue from the *Cockatrice* card game audio suite.
4. `Alert tone, wooden mallet and a bright chime, success sound 1` (Zapsplat Wooden Mallet)	**Set 4** `ZapSplat`	*"You may download and use... credit to 'ZapSplat'... may not... Redistribute..."* **Redistrib: CONDITIONAL** (Packaged app)	`0:02.0` (Fade out at `1.2s`) MP3 (Free) / WAV (Paid)	Warm organic wooden mallet strike paired with a gentle acoustic chime decay at the end of an Orbit session.
5. `Typewriter return bell sound` (#1383) or `Crystal chime` (#3108) (Mixkit Typewriter / Index)	**Set 4** Mixkit	*"...commercial and non-commercial projects for free... can't redistribute... with source files."* **Redistrib: CONDITIONAL** (Closed-source)	`0:01.5` / `0:02.0` WAV / MP3 (`UNVERIFIED` kHz)	Authentic mechanical brass carriage-return bell (#1383) signaling the completion of a study block.

Event 13: Toast / Notification (Subtle)

Target Character: Whisper-quiet, non-startling physical droplet, light glass/brass touch, or muted wooden mallet tap (~100–320 ms) that never interrupts deep study concentration.

Candidate & Direct Link	Set & Author	Quoted Licence & App Redistribution	Duration & Format	One-Line Sound Description
1. drop_002.ogg (8.29 KB, Kenney Interface) or impactGlass_light_001.ogg (Impact)	**Set 1** Kenney Vleugels	*"License: Creative Commons CC0"* **Redistrib: YES** (Unrestricted CC0)	~0.12–0.24s (8.29 KB for drop_002) OGG 44.1 kHz Stereo	Delicate, damped acoustic droplet pop (drop_002) or soft glass tap that registers peripheral awareness quietly.
2. Plop! (0:00.242) or Wet Click (0:00.047) (Freesound Pack #25371)	**Set 2** Breviceps	*"Creative Commons 0 — You can copy, modify, distribute and perform..."* **Redistrib: YES** (CC0)	0:00.242 / 0:00.047 WAV / FLAC (UNVERIFIED kHz)	Soft, natural acoustic droplet plop (242 ms) that provides a gentle, non-electronic toast cue.
3. Notification (pass turn) (Cardsounds.zip) (OGA Card Game Sounds)	**Set 3A** HaelDB	*"License(s): CC0"* **Redistrib: YES** (Unrestricted CC0)	UNVERIFIED (~0.25–0.35s in 238.2 KB ZIP) WAV/OGG (UNVERIFIED kHz)	Purpose-built subtle card-table notification cue for non-blocking state changes.
4. User interface wooden mallet notification tone (Zapsplat Wooden Mallet)	**Set 4** ZapSplat	*"You may download and use... credit to 'ZapSplat'... may not... Redistribute..."* **Redistrib: CONDITIONAL** (Packaged app)	0:02.0 (Trim to single note ~0.30s) MP3 (Free) / WAV (Paid)	Soft, muted single wooden mallet note that feels calm and tactile rather than digital.
5. Dry pop up notification alert (#2356) (Mixkit Notification / Index)	**Set 4** Mixkit	*"...commercial and non-commercial projects for free... can't redistribute... with source files."* **Redistrib: CONDITIONAL** (Closed-source)	0:00.7 WAV / MP3 (UNVERIFIED kHz)	Dry, brief acoustic pop (700 ms) with zero metallic ring or high-frequency harshness.

5. Extended Brief Synthesis: Motion Coupling, M2/M3 Workspaces & Acoustic Spec

5.1 Coupling Audio to Riverline's 31-Moment Motion Catalog & Frequency Rule (Brief §4)

[Judgement] Section 4 of RIVERLINE_BRIEF.md establishes a core architectural principle: *"frequency decides motion (keyboard and very frequent actions are instant); strong ease-out curves; per-moment tempos; one global Motion setting (Instant / Faster / Standard / Reduced); 60 fps."* Audio must obey the exact same frequency hierarchy. If a user holds down ArrowRight to step through 40 hand-history actions in 4 seconds, firing full-length chip cascades on every frame creates harsh comb-filtering and ear fatigue. The table below locks in how the audio engine responds to each global Motion setting and input frequency:

Motion Mode / Input Context	Table Foley Behavior (Events 1–8)	Study & UI Cue Behavior (Events 9–13)	DSP Envelope, Cooldown & Voice Stealing
`Motion: Standard` (Default Desktop / Phone)	Full natural envelope (`100%` duration, 4 round-robin variants per event, subtle `±1.5%` pitch & `±0.8 dB` gain jitter).	Active for explicit clicks/commits (`Confirm`, `Gentle Error`, `Session Finish`, `Toast`); `Soft UI Tick` active on discrete slider stops.	Per-event cooldown `65 ms`; active polyphony cap of 3 simultaneous voices with oldest-voice `15 ms` linear fade-out steal.
`Motion: Faster` (High-Volume Study)	Playback rate scaled to `1.12x` with an early exponential decay window (truncating tails by `25%` for snappy transitions).	`Confirm`, `Gentle Error`, `Session Finish`, and `Toast` active at `1.10x` speed; `Soft UI Tick` attenuated by `-4 dB`.	Per-event cooldown tightened to `50 ms`; polyphony cap of 2 voices so fast street transitions stay crisp and dry.
`Motion: Reduced` (Accessibility / Low Sensory)	Long cascades (`All-In Push`, `Pot Pushed`) replaced by single soft chip/card cues at `-5 dB` attenuation; `Fold` muted.	`Soft UI Tick` muted; `Confirm`, `Gentle Error`, `Toast`, and `Session Finish` active at `-4 dB` with `10 kHz` low-pass softening.	Per-event cooldown widened to `120 ms`; monophonic (1 voice max) so two sounds never overlap.
`Motion: Instant` (Zero-Animation Mode)	**Silent by default** for all street/action transitions (matching zero-duration visual jumps), unless user explicitly toggles *"Keep tactile sound in Instant mode"*.	Only critical boundary cues remain active at `-6 dB`: `Gentle Error` (blocked invalid input) and `Session Finish`.	Zero background buffer scheduling during rapid state jumps; guarantees 0 ms main-thread overhead in `logic.js`.
Rapid Keyboard Scrubbing (`Δt < 90 ms` between keys)	**Auto-Throttled:** When consecutive keyboard steps occur within `< 90 ms`, suppress multi-chip cascades and emit at most one softened `Soft UI Tick` or `cardSlide` every `90 ms`.	All hover/tick sounds suppressed during keyboard repeat (`event.repeat === true`).	Prevents machine-gun comb filtering when scrubbing hands with `ArrowLeft` / `ArrowRight` or stepping through range matrices.

5.2 M2 & M3 Workspace Audio Trigger Matrix & Principle #1–#3 Guardrails

[Judgement] Mapping the 13 audio events onto Riverline's roadmap modules (`RIVERLINE_BRIEF.md §2 & §6`) ensures that audio reinforces physical actions without violating **Principle #1 (Honesty / Confidence Tiers)**, **Principle #2 (Unknown is never fold/zero)**, or **Principle #3 (Heuristic disagreement is never a mistake)**:

Riverline Module (M2 / M3)	Active Audio Events	Suppressed / Silent Actions (Frequency Rule)	Principle #1, #2 & #3 Guardrails (Non-Judgmental Design)
Hand Reconstruction & Table Rules (M2)	`Card Dealt` (assigning hole cards), `Board Card Placed` (flop/turn/river), `Chips Bet` (entering bet sizes), `Confirm` (saving reconstructed hand), `Gentle Error` (selecting an already-dealt card or impossible stack).	Typing numeric stack/pot values into number tiles; tabbing between seat fields.	Leaving a villain's hole cards or action as **Unknown** (Principle #2) triggers `Confirm` (neutral save), never `Gentle Error`.

Riverline Module (M2 / M3)	Active Audio Events	Suppressed / Silent Actions (Frequency Rule)	Principle #1, #2 & #3 Guardrails (Non-Judgmental Design)
Replay & Review (M2)	All 8 table foley events: `Card Dealt`, `Board Card Placed`, `Card Flip` (showdown reveal), `Check Knock`, `Chips Bet (S/M/L)`, `All-In Push`, `Pot Pushed`, `Fold`.	Rapid timeline scrubbing or jumping directly to a street tab (fires only the final landing street's cue).	When stepping onto a decision node where the player's observed play diverged from the baseline, only the physical table action (e.g., `Chips Bet`) plays—**never** an error or warning sound.
Personal Strategy Question Flow (M2)	`Soft UI Tick` (selecting range/sizing buckets via mouse), `Confirm` (locking in a Personal Strategy plan node), `Toast` (strategy branch saved).	Keyboard arrow navigation across the 13×13 hand matrix; hovering over raw data popovers.	**Principle #5:** Committing a Personal Strategy node uses the exact same grounded `Confirm` latch whether it matches or diverges from the heuristic baseline.
Equity, Solver Library & Confidence Tiers (M2 / M3)	`Soft UI Tick` (switching comparison tabs), `Toast` (offline solver computation finished if >1.5s), `Gentle Error` (incompatible board/range syntax).	Hovering hero answer cards or expanding *"calm surface, deep data"* raw breakdowns.	**Principle #1:** Displaying `Exact · price only`, `Verified`, `Estimated`, or `Unavailable` badges is strictly silent—no sonic "upgrade" or "downgrade" cues.
Orbit Sessions, Learn Drills & Achievements (M3)	Table foley during spot playback; `Confirm` when submitting a drill response; `Session Finish (Short, Warm)` when completing an Orbit study block; `Toast` for milestone/achievement unlock.	Advancing to the next spot via keyboard shortcut (uses table deal foley only, no redundant UI click).	**Principle #3 Critical Rule:** When a drill answer is *"inside the baseline's mix"*, *"close"*, or *"outside the baseline's mix"*, Riverline plays the **identical** neutral `Confirm` click. Never play a chime for "inside" or a buzzer for "outside".
Chippy Mascot (M3) & Game Night (M2)	**Chippy (M3):** Uses a dedicated micro-variant of `chip05` (Freesound #817556) or `chipLay1.ogg` (Kenney) attenuated to `-12 dBFS` when presenting a coaching prompt.	Empty-screen illustrations (`Brief §4`) appear silently on view load.	Because Chippy is a *"calm teacher-like poker chip"* (`Brief §4`), his acoustic signature is a single warm clay-chip tap—never a cartoon voice or synthetic beep.

5.3 Acoustic & Temporal Envelope Specification (60 fps Mastering Targets)

[Judgement] To ensure that table foley and study UI cues feel cohesive across multi-hour study sessions, every event is assigned a strict duration ceiling, low-pass softening cutoff (removing fatiguing `>14 kHz` paper/plastic sibilance), calibrated peak level (`-6 dBFS` for primary table actions down to `-14 dBFS` for peripheral ticks and toasts), and minimum re-trigger cooldown floor. Notice how `Gentle Error` is low-passed aggressively at `2.5 kHz` (`-10 dBFS`) so an invalid input registers as a soft physical wooden stop rather than a sharp high-frequency alert.

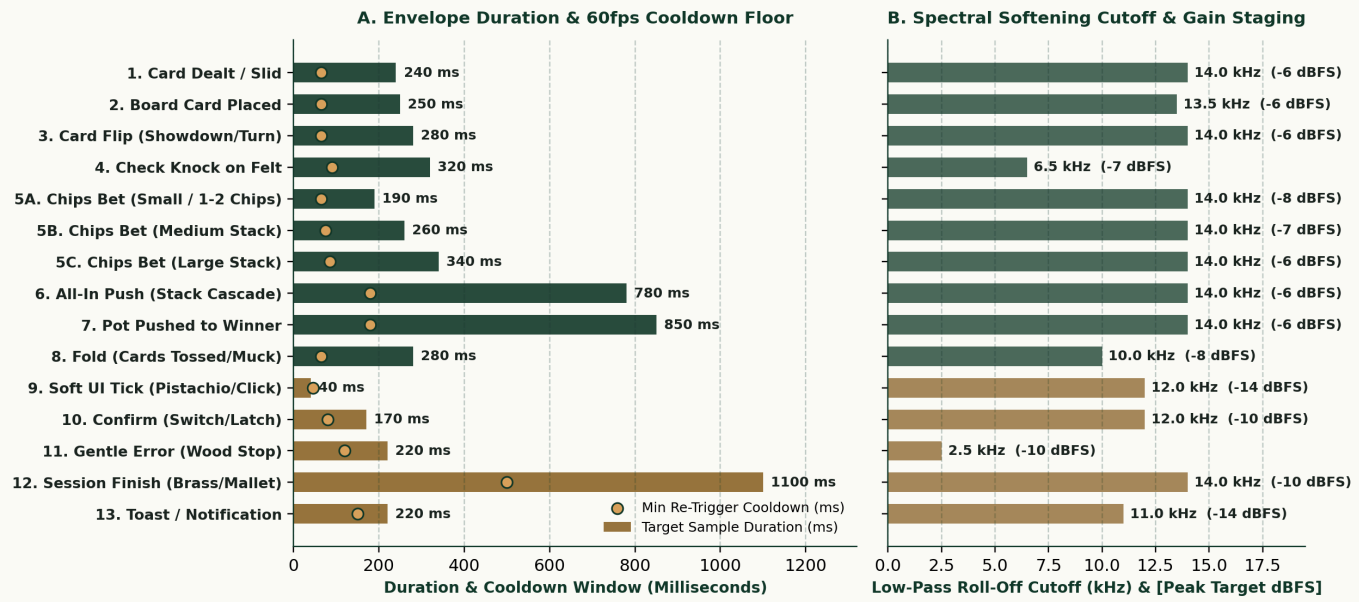


Figure 1: Target duration windows (ms), low-pass softening cutoffs (kHz), normalized peak targets (dBFS), and voice-stealing cooldown thresholds (ms) across all 13 Riverline audio events, aligned with the 60 fps motion catalog in RIVERLINE_BRIEF.md §4.

6. Technical Implementation Blueprint for Riverline (`app/src/core/`)

6.1 Cross-Platform Codec & Container Strategy (`Firefox + Electron + Phone Views`)

[Judgement] Per `RIVERLINE_BRIEF.md §1 & §3`, Riverline runs in **Firefox on Windows** for primary visual/dev checks, ships in a **thin Electron host** (Chromium), and plans a **phone version built on the same code** (which includes iOS Safari / WebKit). While `.ogg` (Vorbis/Opus) works natively in Firefox, Chromium, and Android, iOS Safari's Web Audio `decodeAudioData` has historically lacked OGG Vorbis support on several iOS releases. To guarantee 100% local-first reliability across desktop and phone without bloating the repo:

- **Primary Desktop/Android Format:** `.ogg` (Vorbis or Opus at `48 kHz / 96 kbps`)—all 22 files (including round-robin variants for cards and chips) total **~340 KB**.
- **Universal / iOS Safari Fallback Format:** `.m4a` (AAC-LC at `48 kHz / 128 kbps`) or compact 16-bit `.wav` (all 22 short cues in uncompressed `44.1 kHz` 16-bit Mono WAV total only **~1.45 MB** because 11 of the 13 events are under `0.45 seconds`).
- **[Judgement] Simplest Zero-Transcode Option:** Because the total duration of all 22 audio clips combined is under **9.5 seconds**, shipping `44.1 kHz / 16-bit Mono .wav` (`~1.45 MB` total) or `.ogg + .m4a` dual files (`~720 KB` total) works offline in 100% of browsers (Firefox, Electron, Chrome, iOS Safari) with zero runtime codec branching.

6.2 Offline Mastering & Softening Script (`scripts/master_audio.sh`)

To achieve the required *"real, crisp, lightly softened; card-room foley"* character and eliminate the airy high-frequency hiss noted on raw CC0 takes, run all selected source files once offline through `ffmpeg` before committing them to `app/assets/audio/`:

```
#!/usr/bin/env bash
# Offline mastering pipeline for Riverline M3 Audio (requires ffmpeg)
# 1. Trims leading silence (> -50dB) so playback aligns within <2ms of 60fps animations
# 2. Applies 80Hz high-pass (removes mic stand rumble) and target low-pass ("lightly softened")
# 3. Normalizes peak level to target dBFS (-6dB table foley, -10dB confirm/error, -14dB
tick/toast)
```

```
master_clip() {
  local infile="$1" outfile="$2" max_dur="$3" lpf_hz="$4" peak_db="$5"
  ffmpeg -y -i "$infile" -t "$max_dur" \
    -af
"silenceremove=start_periods=1:start_threshold=-50dB,highpass=f=80,lowpass=f=${lpf_hz},afade=t=ou
t:st=$(awk "BEGIN {print ${max_dur}-0.03}"):d=0.03,volume=${peak_db}dB" \
    -ar 44100 -ac 1 -sample_fmt s16 "$outfile"
}

# Examples:
# master_clip raw/el_boss_571581.wav app/assets/audio/card_slide_1.wav 0.26 14000 -6
# master_clip raw/anthousai_406275.wav app/assets/audio/error_gentle.wav 0.22 2500 -10
# master_clip raw/pistachio_869735.wav app/assets/audio/ui_tick_1.wav 0.04 12000 -14
```

6.3 Zero-Dependency Plain-JS Web Audio Controller (`app/src/core/audio.js`)

Designed for Riverline's plain-JS ES module architecture (`RIVERLINE_BRIEF.md §3`), this controller uses pre-decoded `AudioBuffer` instances (zero synthesis), respects the global `Motion` setting (`Instant / Faster / Standard / Reduced`), enforces per-event cooldowns to prevent keyboard-scrubbing machine-gunning, and never double-signals state:

```
// app/src/core/audio.js — Riverline Real-Foley Web Audio Controller (Zero Dependencies)
const EVENT_CONFIG = {
  cardSlide:     { files: ['card_slide_1.ogg', 'card_slide_2.ogg', 'card_slide_3.ogg'], gain:
0.75, cooldownMs: 65, category: 'table' },
  boardPlace:    { files: ['card_place_1.ogg', 'card_place_2.ogg', 'card_place_3.ogg'], gain:
0.85, cooldownMs: 65, category: 'table' },
  cardFlip:      { files: ['card_flip_1.ogg', 'card_flip_2.ogg'],                       gain:
0.80, cooldownMs: 65, category: 'table' },
  checkKnock:    { files: ['check_knock_1.ogg', 'check_knock_2.ogg'],                   gain:
0.80, cooldownMs: 90, category: 'table' },
  chipsBetSmall: { files: ['chip_small_1.ogg', 'chip_small_2.ogg'],                     gain:
0.75, cooldownMs: 65, category: 'table' },
  chipsBetMed:   { files: ['chip_med_1.ogg', 'chip_med_2.ogg'],                         gain:
0.82, cooldownMs: 75, category: 'table' },
  chipsBetLarge: { files: ['chip_large_1.ogg', 'chip_large_2.ogg'],                     gain:
0.90, cooldownMs: 85, category: 'table' },
  allInPush:     { files: ['all_in_push_1.ogg'],                                       gain:
0.95, cooldownMs: 180, category: 'heavy' },
  potPush:       { files: ['pot_push_1.ogg'],                                         gain:
0.90, cooldownMs: 180, category: 'heavy' },
  foldMuck:      { files: ['fold_muck_1.ogg', 'fold_muck_2.ogg'],                     gain:
0.65, cooldownMs: 65, category: 'table' },
  uiTick:        { files: ['ui_tick_1.ogg', 'ui_tick_2.ogg'],                         gain:
0.35, cooldownMs: 45, category: 'tick'  },
  confirm:       { files: ['confirm_1.ogg'],                                         gain:
0.65, cooldownMs: 80, category: 'ui'    },
  gentleError:   { files: ['error_gentle_1.ogg'],                                    gain:
0.60, cooldownMs: 120, category: 'bound' },
  sessionFinish: { files: ['session_finish_1.ogg'],                                  gain:
0.75, cooldownMs: 500, category: 'bound' },
  toast:         { files: ['toast_subtle_1.ogg'],                                   gain:
0.45, cooldownMs: 150, category: 'ui'    }
};

export function createAudioEngine({ getMotionMode = () => 'Standard', basePath =
'./assets/audio/' } = {}) {
  let ctx = null;
  const buffers = new Map();
  const lastPlayedAt = new Map();
```

```
  const roundRobinIdx = new Map();

  function ensureContext() {
    if (!ctx && typeof window !== 'undefined') {
      const AudioCtx = window.AudioContext || window.webkitAudioContext;
      if (AudioCtx) ctx = new AudioCtx();
    }
    if (ctx && ctx.state === 'suspended') ctx.resume().catch(() => {});
    return ctx;
  }

  async function preloadAll() {
    const audioCtx = ensureContext();
    if (!audioCtx) return;
    const uniqueFiles = new Set(Object.values(EVENT_CONFIG).flatMap(c => c.files));
    await Promise.all([...uniqueFiles].map(async (file) => {
      try {
        const res = await fetch(basePath + file);
        const arrayBuf = await res.arrayBuffer();
        buffers.set(file, await audioCtx.decodeAudioData(arrayBuf));
      } catch (_) { /* Missing or uncached asset fails silently per local-first resilience */ }
    }));
  }

  function play(eventName, { isKeyboardRepeat = false } = {}) {
    const cfg = EVENT_CONFIG[eventName];
    if (!cfg || isKeyboardRepeat) return;
    const motion = getMotionMode(); // 'Instant' | 'Faster' | 'Standard' | 'Reduced'
    if (motion === 'Instant' && cfg.category !== 'bound') return;
    if (motion === 'Reduced' && cfg.category === 'tick') return;

    const now = performance.now();
    const minGap = motion === 'Reduced' ? Math.max(120, cfg.cooldownMs) : cfg.cooldownMs;
    if (now - (lastPlayedAt.get(eventName) || 0) < minGap) return;
    lastPlayedAt.set(eventName, now);

    const audioCtx = ensureContext();
    if (!audioCtx) return;

    // In Reduced motion, downgrade heavy chip cascades to single small chip cue
    const effectiveCfg = (motion === 'Reduced' && cfg.category === 'heavy') ?
EVENT_CONFIG.chipsBetSmall : cfg;
    const idx = (roundRobinIdx.get(eventName) || 0) % effectiveCfg.files.length;
    roundRobinIdx.set(eventName, idx + 1);
    const buf = buffers.get(effectiveCfg.files[idx]);
    if (!buf) return;

    const src = audioCtx.createBufferSource();
    const gainNode = audioCtx.createGain();
    src.buffer = buf;
    src.playbackRate.value = (motion === 'Faster' ? 1.12 : 1.0) * (1 + (Math.random() - 0.5) *
0.03);
    const modeScalar = motion === 'Reduced' ? 0.55 : 1.0;
    gainNode.gain.value = effectiveCfg.gain * modeScalar;

    src.connect(gainNode).connect(audioCtx.destination);
    src.start(0);
  }
```

```
  return { preloadAll, play };
}
```

7. Risks, Cost/Effort Summary & Open Decisions for the Owner

Dimension	Cost / Effort / Risk	Details & Mitigation
Direct Licensing Cost	**$0.00** (Free)	100% of candidates in **Set 1 (Kenney CC0)**, **Set 2 (Freesound CC0 / CC-BY)**, and **Set 3A (OpenGameArt CC0)** are free for commercial and non-commercial use with zero royalties.
Engineering Effort	**~3–4 Hours Total**	~1.5 hours to download, slice multi-take WAVs, and run `master_audio.sh`; ~2 hours to wire `app/src/core/audio.js` into `logic.js` action dispatchers and add a Node DOM harness unit test verifying silence under `Motion: Instant`.
Open-Source Repo Risk	**ZERO** (if Sets 1, 2, 3A) / **HIGH** (if Sets 3B, 4)	**[Judgement]** Avoid committing Sonniss GDC, Zapsplat, Mixkit, or Pixabay files into Git if Riverline may ever open-source its repository. Stick to Kenney CC0, Freesound CC0/CC-BY, and OpenGameArt CC0.
BBC RemArc Risk	**DISQUALIFIED**	Flagged per prompt instructions: BBC RemArc Clauses 7 & 8 strictly prohibit commercial use or paid/supporter tiers without commercial licensing via Pro Sound Effects.

Open Questions for Owner Sign-Off

1. **CC0-Only vs. CC-BY Attribution Line:** Every event in Section 4 has 2–4 pure `CC0` candidates requiring zero attribution. Do you want to enforce a strict **CC0-only policy** across the repository, or are you comfortable including a 2-line credit in Riverline's `About / Licences` view to unlock the two `CC-BY` Freesound recordings (f4ngy's Card Flip under CC-BY 4.0 and Benjamin152's Zoom H4n Poker Chips & Wood under CC-BY 3.0)?
2. **Recommended Hybrid Palette (Set 1 + Set 2): [Judgement]** Rather than using Set 1 or Set 2 in isolation, the highest-quality zero-cost configuration is a **Set 1 + Set 2 Hybrid** (100% CC0/CC-BY): use **Set 2 (Freesound)** for close-mic clay chips (SilverDubloons, farthestar), finger check-knock (Churrumaki), pistachio-shell UI ticks (SignatureSoundsOrg), C414 switch/latch confirm (loganzsound), and RX-denoised warm brass chimes (loganzsound), while using **Set 1 (Kenney Casino Audio)** for round-robin `cardSlide1–4`, `cardPlace1–4`, and `cardShove1–4`.